Framework for Secure and Sustainable Management of Legal Documents in Academic Libraries Using Smart Contracts

Deepak Bhatt[1*] and Amar Jeet Rawat [2]

[1] Uttaranchal University, Dehradun, INDIA
[2] Uttaranchal University, Dehradun, INDIA
Email: bhattdpac@gmail.com

Abstract. In academic libraries, the digitization of legal documents poses significant long-term preservation, security, and access concerns. This paper proposes a new architecture to overcome these concerns, which considers the solution in terms of a Hybrid Intelligence (HI) system---a system that supplements human intelligence without replacing it. Our framework establishes an environment for librarians and AI components to work collaboratively by incorporating blockchain, smart contracts and Natural Language Processing (NLP). In our framework, we leverage the immutability of a permissioned blockchain (Hyperledger Fabric) and decentralized storage (IPFS) to create a tamper-proof repository, while document lifecycle policies are automated and enforced using smart contracts. Emphasizing responsible governance and explainability, the HI approach addresses known issues with blockchain adoption. In this paper, I will highlight the alignment of the system with the principles of Collaborative, Responsible and Explainable AI, outline the architecture of the system, and detail a use case in a university library context. The proposed framework presents a realistic pathway for developing secure, efficient, and sustainable digital preservation systems for sensitive legal documents

Keywords: Blockchain, Smart Contracts, Hybrid Intelligence, Digital Preservation, Legal Tech, Explainable AI.

1 Introduction

Academic libraries contain very large collections of legal documents, including theses, dissertations, and institutional documents. The change to a technology-driven environment has made the management and preservation of these sensitive resources more complicated, and continuing to ensure the long term authenticity, integrity, and usability of digital legal documents is a complex and interdisciplinary challenge, with related issues of technological obsolescence, media fragility, and limited, enforceable policies [4]. Centralized database systems jeopardize the integrity of the academic and legal documentary record and access by scholars, and put data at risk of loss, disenfranchised editing, and censorship [5]. Managing rights to access and ensuring that management

R. Tanwar et al. (eds.), *Proceedings of 3rd International Conference of Library & Technology on Artificial Intelligence and Humanities in Library and Education 4.0 (AIHLE 2025)*, Advances in Intelligent Systems Research 205,
https://doi.org/10.2991/978-94-6239-618-0_18

still meets ethical and legal practices is also a manual process that is not always straightforward [6].

The growth of blockchain technology represents an innovative solution to the problems described above. A blockchain is a decentralized, irreversible record, an immutable secure ledger of transactions and a method of managing information [1]. Furthermore, smart contracts (self-executing contracts written in code) provide the ability to automate complicated business workflows and enforce previously agreed-upon terms without a third party [2, 7]. This article introduces a tailored framework to better manage legal documents in academic libraries: a Hybrid Intelligence(HI) system [12]. Hybrid Intelligence describes the concept whereby human and machine intelligence can work together to boost human intelligence rather than to replace it. Our aim is not a fully automated service in the management of legal documents, although an electronic environment can be created where librarians, researchers, and elements of artificial intelligence interacted together. In particular, this will provide a means, through the application of blockchain, smart contracts, and NLP, to build a system that enhances the sustainable digital preservation of legal documents while also adding value for researchers and the public. This paper offers a practical and actionable use case in the fast-emerging area of hybrid intelligence and can be easily accommodated into the AIHLE 2025 themes of "Emerging Technology Applications in Libraries," "Sustainable Digital Preservation," and "AI and Machine Learning in Library Systems and Services."

The main contributions of this paper are as follows: 1. We propose a novel framework for the secure and sustainable management of legal documents in academic libraries, framed as a Hybrid Intelligence(HI) system that augments human intellect.

2. We detail a system architecture that integrates a permissioned blockchain (Hyperledger Fabric) decentralized storage (IPFS), and a Natural Language Processing (NLP) module to ensure data integrity, security, and accessibility.

3. We align the proposed framework with the core principles of collaborative, responsible, and explainable AI, providing a practical governance model for a real-world academic setting.

4. We present a detailed illustrative use case for managing the lifecycle of a PhD thesis, demonstrating the practical application and benefits of the framework.

2 Related Work

The proposed framework is situated at the nexus of three rapidly advancing fields: the automation of smart contracts, blockchain architecture, and the application of Artificial Intelligence (AI) in decentralized systems. While our focus is specifically academic libraries, there are examples of related blockchain-based frameworks being considered, such as for the management of lawful evidence in digital forensics [16], indicating the potential for application in other domains.

2.1 Smart Contract Generation and NLP

The automated creation of smart contracts from legal text represents an important area of research. There is a preliminary contribution by Ahmed et al. [2], who developed a prototype system that generates smart contracts from legal agreements. Their work consists of a custom Named Entity Recognition (NER) model that identifies important features in legal rental agreements with 96 percent accuracy, together with a custom parser for date formats that allows your model to achieve 96 percent performance and exceed standard libraries. Their approach then uses a template approach to generate code in Solidity. While a promising approach, there was a specificity in the type of document (rental agreements) and support for a single blockchain implementation (Ethereum). Our work builds on this approach to propose a flexible and modular architecture approach that would support a larger range of legal documents in academic libraries, and we develop the architecture for a different blockchain technology (Hyperledger Fabric), which is more suitable from an enterprise perspective. A related study by Monteiro et al. [8] proposes generating smart contract skeletons, which adds to the body of literature demonstrating the ability of NLP to bridge the gap between legal language and executable code.

2.2 Smart Contract Architecture and Legal Frameworks

Here it is important to understand the architecture behind smart contracts to know how to build safe and reliable applications. The literature has traversed the modern landscape in breadth, with examples including that of Mazlan et al. [3] who provided a comprehensive review of smart contract architectures, usage across many industries, and security and legal issues with these uses. In analyzing the literature, the authors determined that notwithstanding the utility and benefits of smart contract architecture, features (decentralized, automated, secure, and transparent via an immutable record) that are [3] useful for applications, there still remain challenges. Specifically, Mazlan et al. identified challenges with scalability, energy demand, and institutional regulatory compliance as areas for concern [9]. We respond to those challenges with our permission blockchain solution (Hyperledger Fabric) which has facilitated better scalability and control in business settings [10]. Furthermore, we developed the smart contracts from the legal and ethical perspective of an academic library, as cited in the literature. Mazlan et al. wrote, "By way of these innovations, smart
contracts continuously open the door to a revolution for the decentralized digital economy that will affect productivity and trust" [3], and that is part of what we are trying to do.

2.3 The Convergence of Blockchain and AI

The combination of AI and blockchain is an impactful paradigm that could establish a new type of intelligent system that is decentralized. A thorough survey by Salah et al. [1] outlines this area well, explaining how blockchain can provide AI with secure, trusted and auditable data. They assert, "the combination of AI and blockchain can create a secure, immutable, decentralized system for the very sensitive data needed to be collected, stored, and utilized in AI driven systems" [1]. This is directly relevant to our proposed framework which employs blockchain as a secure system to apply NLP when analyzing and managing currently sensitive legal documents. The survey of Salah et al. also notes several challenges for future research, including the need for AI specific consensus protocols and robust governance models. While the current work does not explore new consensus protocols, it provides a practical use case for this nascent field by illustrating how a permissioned blockchain can be used as the secure data layer for this NLP application, which may be a real-world case study. This application contributes to the literature as much of the research focuses on theoretical integrations rather than practical applications.

2.4 Challenges to Blockchain Adoption in Academia

Academia offers a considerable opportunity for blockchain, but recognizing issues and barriers to adoption is paramount. Kosmarski [11] has produced a detailed critical review of these issues based on qualitative research with to academic stakeholders, including librarians. The two biggest issues include:
1. Usability and User Experience: Many current blockchain-based applications do not feature user friendly interfaces and require technical expertise, making it difficult for the wider academic community to use these technologies.
2. Legality and Uncertainty in Regulation: The legality of smart contracts and cryptocurrencies appears to remain diluted depending on jurisdictions, creating environments in which it is hazardous for institutions to invest in and utilize blockchain technologies.
3. Conflict of Values: The emergence of token-based economies and market-oriented incentives can be in conflict with traditional values of open inquiry and collaboration in scholarship.
4. Governance and Power Dynamics: Governance of blockchain-based systems introduces complex political and ethical issues surrounding the regulatory environment, protocol development, and data control. Additionally, in the case of DAOs in empirical research, voting power is frequently structured to be held with a small group of information silos, thereby constructing only the "illusion of decentralization" [13].

These quandaries highlight the important consideration of addressing blockchain-assisted systems within higher education at the level of design and implementation, in a deliberate and intentional manner. Our project also provides further support for the interoperability protocols proposed in emerging literature, such as the Blockchain Academic Credential Interoperability Protocol (BACIP) [15], which aims to establish universal standards for academic credentials. In being intentional about a specific use case, incorporating governance and security features by utilizing a permissioned blockchain, and drawing from the context of the needs of the academic library community, the framework asserted in this article may work to mitigate some of these dilemmas.

Table 1. Blockchain Adoption Challenges and Framework Responses

Challenge	Description	Response
Usability [1]	**High technical barrier for non-experts**	Abstracts complexity via a user-friendly Web Portal
Legal & Regulatory [4]	**Unclear legal status of smart contracts**	Operates within a consortium, simplifying compliance
Value Conflict [6]	**Token-based economies conflict with academic values**	Avoids tokenization, focusing on utility
Governance & Power [9]	**Public DAOs suffer from power concentration**	Uses a permissioned model (Hyperledger Fabric) to ensure accountability

3. Methodology

This study employs Design Science Research (DSR) methodology, a problem-solving approach that focuses on creating new artifacts to solve problems in the world. The artifact in question, for example, is the proposed framework for managing legal documents within academic libraries. The specific DSR process we followed has the following phases:

3.1 Problem Identification and Motivation

We started by identifying the digital preservation and legal document management issues in academic libraries, as described in the Introduction.

3.2 Definition of Objectives for a Solution

After identifying the issues, we then defined the objectives for our solution, which was mainly to improve security, transparency, efficiency, and sustainability.

3.3 Design and Development

We developed and designed the proposed framework using the three elements in 3.2 incorporating NLP, Blockchain, and Smart Contracts.

3.4 Demonstration

The assessment of the framework will depend on its ability to satisfy the identified objectives, and the assessment will include both a qualitative review of the framework features and a discussion of the intended impact in this review.

4. The Proposed Framework

In response to the digital preservation and legal document management challenges in the academic library environment, we offer a framework that integrates Natural Language Processing (NLP), a blockchain, and smart contracts in a secure, transparent, and efficient manner for the lifecycle of a legal document.

a. System Architecture

The architecture of the suggested framework consists of four primary levels: The Application Layer, the NLP Service Layer, the Blockchain Layer, and the Decentralized Storage Layer.

1. Application Layer: This is the user-facing portal for librarians, researchers, and all other interested parties to access the system. Through this layer, users can upload documents, find documents in the repository and manage access approval.
2. NLP Service Layer: This layer has the role of intelligently analyzing all legal documents. When a document is uploaded, it is sent to this layer for processing. Key metadata is extracted, entities are identified, and a structured representation of the content is generated by the NLP pipeline.
3. Blockchain Layer: Built on Hyperledger Fabric, this is the core of the framework. It keeps an immutable ledger of all document related transactions, including records of submission, metadata, access permissions, and modifications. The smart contracts governing the system are deployed on this layer.
4. Decentralized Storage Layer: The actual files are stored off-chain in a decentralized system like the InterPlanetary File System (IPFS) to ensure long-term preservation and availability. To ensure data integrity while minimizing blockchain storage load, only the document's hash is stored on-chain.

Table 2. System Architecture Layers and Components

Layer	Components	Function
Application	**Web Portal / UI**	User-facing interface for document interaction
NLP Service	**Metadata Extraction**	Intelligent analysis of documents for metadata generation
Blockchain	**Hyperledger Fabric, Smart Contracts**	Maintains immutable ledger; automates and enforces rules
Storage	**IPFS**	Securely stores encrypted documents off-chain

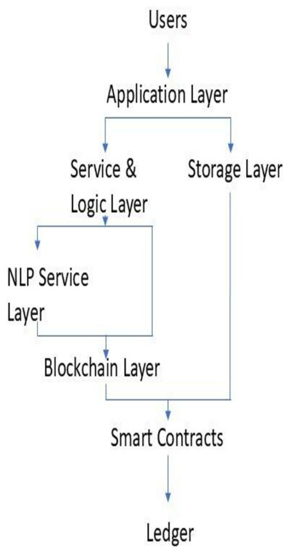

Fig. 1 System Architecture Diagram

Fig. 1 System Architecture Diagram

b. NLP Module for Legal Document Analysis

The NLP module is a critical component for making legal documents discoverable and manageable. Its process is as follows:

1. Document Ingestion: Through the web portal, a user uploads a legal document (e.g., PDF, DOCX).
2. Text Extraction and Preprocessing: From the document, raw text is extracted. It is then cleaned and normalized for analysis.
3. Named Entity Recognition (NER): To identify key entities, a custom-trained NER model, similar to the method of Ahmed et al. [2], is used, such as:

   Parties: Names of individuals or organizations. Dates: Effective dates, expiration dates, etc.

   Jurisdiction: The legal jurisdiction of the document. Obligations and Rights: Key clauses defining party rights and obligations.

4. Metadata Generation: A rich set of metadata for the document is generated from the extracted entities. This metadata is then used for indexing in a searchable database, with a summary stored on the blockchain.

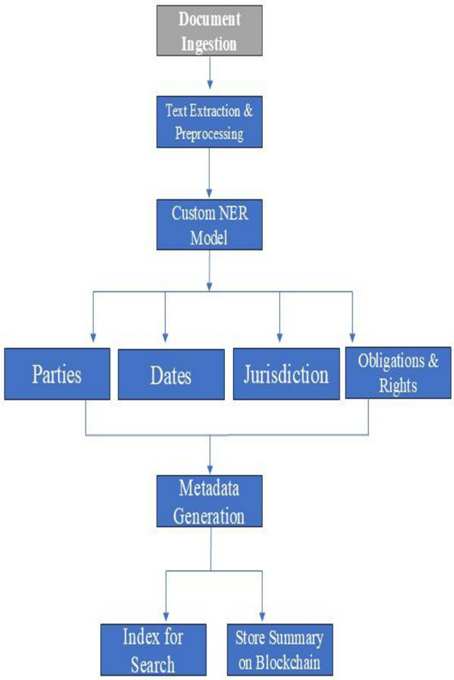

Fig. 2 NLP Metadata Extraction Process

Fig. 2 NLP Metadata Extraction Process

c. Blockchain and Smart Contract Design

For the blockchain layer, we selected Hyperledger Fabric because its permissioned nature is ideal for an enterprise environment like a university library system [10]. This choice is consistent with other academic and accreditation-focused blockchain frameworks, which also leverage Hyperledger Fabric for its modularity and permissioned architecture [18]. The main components are:

1. Channels: To ensure only authorized participants (e.g., library staff, legal department) can view and transact, a private channel is created.
2. Smart Contracts (Chaincode): A suite of smart contracts, developed in Go, governs the document lifecycle:
a. `DocumentLifecycle.go`: Manages submission, verification, and archival,

recording the IPFS hash and metadata.
b. `AccessControl.go`: Manages document permissions with granular control.
c. `AuditTrail.go`: Provides an immutable audit trail for all actions, ensuring transparency.

d. Digital Preservation and Security Model

The framework uses a multi-pronged approach to ensure long term preservation and security:
1. Irreversibility: By storing the document's hash on the blockchain, we can verify the integrity of the document. Tampering with the document off-chain
will change the hash, signaling an integrity issue with the document.
2. Decentralization: By storing documents on IPFS, we eliminate a single point of failure from a centralized server, ensuring availability.
3. Encryption: All documents are encrypted before uploading to IPFS. The `AccessControl` smart contract is responsible for decryption keys and ensures access is only granted to identified parties.
4. Permissioned Access: By design, Hyperledger Fabric has a permissioned network in which only trusted parties can access the network, which prevents unauthorized access to the ledger.
The primary challenge of every blockchain network interacting with off-chain data is the cost and complexity of verification. While storing a hash on-chain is a relatively low-cost transaction, complicating the verification of off-chain data can be cost-prohibitive. To resolve this concern, our framework can be enhanced by using a "validation-on-demand" approach based on the Testimonium scheme [14]. In this approach, we accept data optimistically and resolve disputes if the data is invalid. This low trust method relies on an incentive structure for "disputers" which can significantly decrease on-chain cost verification for the system, resulting in a more cost-effective and scalable system.

e. A Hybrid Intelligence Perspective

The proposed framework is best understood as a Hybrid Intelligence (HI) system, designed to augment human intellect rather than replace it [12]. This aligns with the growing recognition of the powerful synergy between AI and blockchain, where AI can enhance the security, scalability, and efficiency of blockchain systems [17]. For the system's successful adoption and ethical implementation, this perspective is vital. Our framework can be analyzed through the four key challenges of HI:
1. Collaborative HI: The framework is inherently collaborative. Acting as an intelligent assistant to the librarian, who maintains ultimate authority, is the NLP module. The system is used by researchers to discover and access documents. As impartial administrators of community-agreed rules, the smart contract's function.
2. Adaptive HI: Although the current framework possesses limited adaptive features, it is designed for extensibility. Future work will concentrate on integrating machine learning algorithms that enable the system to learn from user interactions. For instance, from

the corrections made by librarians, the NLP module could learn to improve its accuracy over time.
3. Responsible HI: A core design principle of the framework is responsibility. A key design choice that bolsters responsibility is the use of a permissioned
blockchain (Hyperledger Fabric). A more controlled and accountable governance structure is permitted by a permissioned system, unlike public blockchains where governance can be prone to power concentration and the "illusion of decentralization" [13]. Participants in our framework are known and trusted members of the academic community, and the institution can design and enforce the system's rules, ensuring alignment with library policies and ethical guidelines. Accountability and responsible governance are prioritized by this approach over the ideal of complete, uncontrolled decentralization
4. Explainable HI (XAI): The framework is designed with explainability as a priority. For example, when metadata is extracted by the NLP module, a reference to the specific passage in the document from which the information was extracted can also be provided. This enables librarians to quickly verify the accuracy of the AI's output [11].

5. Illustrative Use Case: Managing PhD Theses in a University Library

To demonstrate the framework's practical application, we consider the lifecycle of a law PhD thesis within a university's digital library.
1. Submission: A PhD candidate in intellectual property law uploads their final thesis, "The Role of Blockchain in Copyright Management," to the library's secure portal. The document is processed by the NLP module, which extracts not only the author, title, and abstract, but also key legal terms (e.g., "fair use"), cited cases, and statutory references. The thesis is then encrypted and stored on IPFS. On the blockchain, a new transaction is created, containing the IPFS hash, the rich metadata, and an initial "Pending Approval" status.
2. Verification: A subject librarian with legal expertise is notified of the submission. On their dashboard, they review the thesis and the AI-generated metadata. The librarian confirms the entity accuracy and, with a single click, invokes a function in the `DocumentLifecycle` smart contract to set the status to "Approved." This action, digitally signed by the librarian, is recorded as an immutable and attributable transaction on the blockchain [12].
3. Access by a Researcher: Years later, a researcher from a partner institution, conducting a literature review on blockchain's legal implications, discovers the thesis via the portal's detailed metadata. They request full text access. The `AccessControl` smart contract automatically consults the university's encoded access policies. For instance, the policy might grant immediate access to partners while requiring a formal request from external researchers. Here, a temporary, time-limited decryption key is granted by the smart contract, allowing the researcher to view the thesis. This access event, including the researcher's affiliation and access duration, is recorded on the blockchain, providing a transparent, granular audit trail [13].

4. Long-Term Preservation: Decades later, local servers may have been replaced, and file formats evolved. However, on the decentralized IPFS network, a copy of the thesis remains securely preserved and
accessible. The complete, unbroken history of the document, from submission and verification to every access event, is contained on the blockchain, which has been maintained by a consortium of academic libraries. For future generations of legal scholars, this ensures its provenance and authenticity, directly tackling the critical challenges of technological obsolescence and data integrity [14].

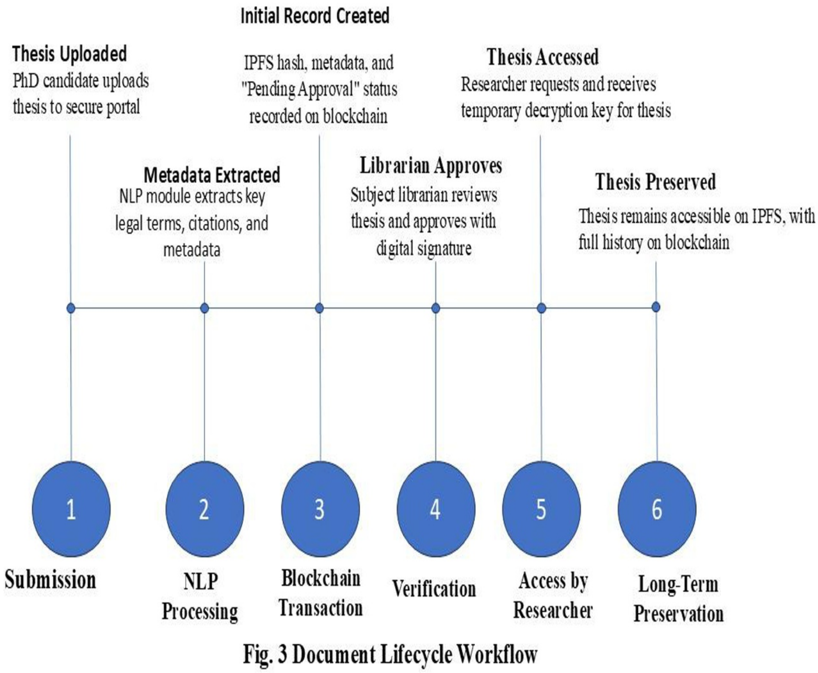

Fig. 3 Document Lifecycle Workflow

6. Evaluation

The evaluation of the framework centers on its ability to meet the core objectives of the DSR methodology: enhancing security, transparency, efficiency, and sustainability in legal document management. A qualitative evaluation highlights the potential benefits:

Table 3. Evaluation Objectives and Key Framework Features

Objective	Key Framework Features
Security	**Permissioned blockchain (Hyperledger Fabric), encryption of documents on IPFS**
Transparency	**Immutable ledger recording all transactions, providing a complete and auditable trail**
Efficiency	**Automation of metadata extraction (NLP) and access control (Smart Contracts)**
Sustainability	**Use of decentralized storage (IPFS) to mitigate risks of technological obsolescence**

While a full-scale quantitative evaluation is outside this paper's scope, the qualitative analysis shows the framework's potential to solve key digital preservation challenges in academic libraries.

a. Theoretical Performance and Scalability Analysis

Although a physical prototype was never built, a theoretical evaluation of the components selected for the tracking device indicates favorable performance characteristics. However, it should be noted that Hyperledger Fabric is a permissioned distributed ledger technology that does not utilize the computationally intensive proof-of-work consensus algorithm used by public blockchain technologies, such as Bitcoin. Rather, Hyperledger Fabric uses a differentiated consensus protocol (for example, Raft) that is designed specifically for enterprise contexts where high throughput and low latency are required for the business environment. Furthermore, our on chain/off-chain data model significantly enhances scalability [15].

By only storing immutable hashes and metadata on-chain and keeping the large legal documents in a dedicated off-chain repository (IPFS), we avoid bloating the blockchain ledger. This design ensures that the transaction throughput remains high and is not bottlenecked by the storage and processing of large files, making the framework scalable to a large number of documents and users [16].

Furthermore, the significant challenges to blockchain adoption in academia, as identified by Kosmarski [11], are taken into account by the proposed framework's design. We address the threat of governance and power relations present in public, permission less systems by leveraging a permissioned blockchain (Hyperledger Fabric). Our insistence on a discrete and pragmatic use case in a library context -- having chosen not to tokenize the entire scientific process -- is a requisite response to the "conflict of values" problem. Our goal is to proffer an instrument for librarians and researchers to use, not a wholesale change to academic incentives [17].

7. Conclusion

This article sets forth an innovative framework for the sustainable digital preservation and secure management of legal documents in academic libraries. We have constructed our solution as a Hybrid Intelligence (HI) system meant to enhance librarians and researchers, rather than replace them. By integrating blockchain technology, smart contracts, and Natural Language Processing, the proposed framework addresses key challenges of data integrity, security, and accessibility, while also aligning with the core principles of collaborative, responsible, and explainable AI [12].

The qualitative evaluation of the framework highlighted its potential to enhance the security, transparency, efficiency, and sustainability of digital preservation processes. The use of a permissioned blockchain, decentralized storage, and automated workflows provides a robust and reliable solution for managing sensitive legal documents. Acknowledging the broader challenges of blockchain adoption in academia [11], this work represents a pragmatic step towards realizing the benefits of this technology in a specific, controlled environment and contributes a practical use case to the growing field of Hybrid Intelligence.

We must also recognize the limitations of this work. The proposed framework is, at present, conceptual. A full implementation would require significant engineering effort, especially in developing and training the NLP module, which needs a substantial corpus of legal documents to achieve high accuracy. Moreover, the economic and social incentives of the proposed "validation-on-demand" pattern would need to be carefully designed and tested to ensure system security and reliability.

Future work will focus on the framework's implementation and quantitative evaluation in a real-world academic library setting. This will involve developing a fully functional prototype and conducting user studies to assess its usability and effectiveness. Further research will also explore the integration of more advanced AI techniques for legal document analysis, the development of adaptive capabilities, and the enhancement of the framework's explainability, in line with the HI research agenda [12]. Additionally, we will investigate alternative governance models for the framework, aiming to balance the benefits of decentralization with the need for accountability and effective decision-making [13]. Finally, we will explore the implementation of cost-efficient verification mechanisms, such as the "validation-on-demand" pattern proposed by Testimonium [14], to enhance the framework's scalability and economic viability.

References

1. K. Salah, M. H. U. Rehman, N. Nizamuddin, and A. Al-Fuqaha, "Blockchain for AI: Review and open research challenges," *IEEE Access*, vol. 7, pp. 10127-10149, 2019.
2. S. U. Ahmed, A. Danish, N. Ahmad, and T. Ahmad, "Smart contract generation through NLP and blockchain for legal documents," *Procedia Computer Science*, vol. 235, pp. 2529-2537, 2024.

3. F. Mazlan, N. F. Omar, N. N. M. S. N. M. Kamal, and A. A. Zainuddin, "Comprehensive insights into smart contracts: Architecture, sectoral applications, security analysis, and legal frameworks," *Journal of Informatics and Web Engineering*, vol. 4, no. 1, 2025.
4. P. Conway, "Preservation in the age of Google: Digitization, digital preservation, and dilemmas," *The Library Quarterly*, vol. 80, no. 1, pp. 61-79, 2010. [5] D. S. H. Rosenthal, "The challenge of digital preservation," *Queue*, vol. 13, no. 10, pp. 10-19, 2015.
5. R. Savolainen, "Information sharing and information avoidance: A qualitative study of the information behavior of a hobby community," *Journal of the Association for Information Science and Technology*, vol. 68, no. 4, pp. 903-914, 2017.
6. Z. Zheng, S. Xie, H. Dai, X. Chen, and H. Wang, "Blockchain challenges and opportunities: A survey," *International Journal of Web and Grid Services*, vol. 14, no. 4, pp. 352-375, 2018.
7. E. Monteiro, R. Righi, C. da Costa, and D. Singh, "Combining Natural Language Processing and Blockchain for Smart Contract Generation in the Accounting and Legal Field," in *Intelligent Human Computer Interaction*, 2021, pp. 307-321.
8. K. Christidis and M. Devetsikiotis, "Blockchains and smart contracts for the internet of things," *IEEE Access*, vol. 4, pp. 2292-2303, 2016.
9. E. Androulaki et al., "Hyperledger fabric: a distributed operating system for permissioned blockchains," in *Proceedings of the thirteenth EuroSys conference*, 2018, p. 30.
10. Kosmarski, "Blockchain adoption in academia: Promises and challenges," *Journal of Open Innovation: Technology, Market, and Complexity*, vol. 6, no. 4, p. 117, 2020.
11. Z. Akata et al., "A research agenda for hybrid intelligence: Augmenting human intellect with collaborative, adaptive, responsible, and explainable artificial intelligence," *Computer*, vol. 53, no. 8, pp. 18-28, 2020.
12. R. Fritsch, M. Müller, and R. Wattenhofer, "Analyzing voting power in decentralized governance: Who controls DAOs?," *Blockchain: Research and Applications*, p. 100208, 2024.
13. P. Frauenthaler, M. Sigwart, C. Spanring, and S. Schulte, "Testimonium: A cost efficient blockchain relay," *arXiv preprint arXiv:2002.12837*, 2020. [15] J. A. B. Moya, "Blockchain for Academic Integrity: Developing the Blockchain Academic Credential Interoperability Protocol (BACIP)," *arXiv preprint arXiv:2406.15482v1*, 2024.
14. M. Li, C. Lal, M. Conti, and D. Hu, "LEChain: A blockchain-based lawful evidence management scheme for digital forensics," *Future Generation Computer Systems*, vol. 115, pp. 406-420, 2021.
15. B. Xing and T. Marwala, "The Synergy of Blockchain and Artificial Intelligence," *SSRN Electronic Journal*, 2018. [Online]. Available: https://ssrn.com/abstract=3225357
16. Kumar, Y. and Kumar, N.: Electro-optical Effect of 4-n-alkyl-sulfanyl-4' isothiocyanate-biphenyl Liquid Crystal Homologous Series Under Terahertz Frequency: A Theoretical Approach. *Makara Journal of Science,* 27(4), 255-263 (2023)
17. Y. Luo, et al., "A Secure Storage and Verification Framework Based on Consortium Blockchain for Engineering Education Accreditation Data," *Computers, Materials & Continua*, 2025